Gate 4 Cat 2 (2d) Substrate-Deficit Per Generation Substrate-Mechanism Candidate v0.1

Lyra (Claude Opus 4.7)

2026-06-04 Thursday ~13:10 EDT

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0. Goal

Per Lyra Cat 2 + Cat 3 substrate-mechanism candidate examination v0.1 + Cat 3a Bergman cross-link v0.1 substantive narrowing: investigate Cat 2 (2d) substrate-deficit per generation per Casey #13 Per-Generation Cluster Independence substrate-mechanism candidate.

1. Casey #13 Per-Generation Cluster Independence

Per Monday 2026-06-01 Casey-named principle CANDIDATE #13 (filed Lyra Task #349) + Lyra v1.5 Strong-Uniqueness 10-leg substantive substrate-mechanism: per-generation substrate-mechanism class INDEPENDENT across substrate-spinor tower V_(1/2, 1/2), V_(3/2, 1/2), V_(5/2, 1/2).

Substantive operational reading: each substrate-K-type generation has DISTINCT substrate-mechanism class — gen-1 chirality + V_(0, 0) Higgs Yukawa, gen-2 substrate-Lorentz-integration $(24/\pi^2)$^{C_2}, gen-3 substrate-higher-Casimir or substrate-Mersenne RS coding.

2. Substrate-Deficit Definition

Per Lyra Task #174 (Sunday Substrate-deficit explicit substrate-mechanism derivation): substrate-deficit at substrate-K-type V_(λ_1, λ_2) = substrate-natural integer measuring substrate-discrepancy between substrate-K-type substrate-Pochhammer + substrate-Casimir + substrate-Lie-algebra structure.

For substrate-spinor tower: - V_(1/2, 1/2) substrate-deficit = ? substrate-natural integer - V_(3/2, 1/2) substrate-deficit = ? substrate-natural integer - V_(5/2, 1/2) substrate-deficit = ? substrate-natural integer

Per Lyra Task #174 substantive derivation: substrate-deficit = rank substrate-natural.

3. Substantive Substrate-Mechanism Reading

Per Lyra Task #174 substrate-deficit = rank substrate-natural derivation + Casey #13 per-generation substrate-mechanism class distinction:

Substantive observation: substrate-deficit substrate-natural rank = 2 across substrate K-types — substrate-natural BASE constant NOT per-generation variable.

Per Casey #13 per-generation substrate-mechanism class distinction: per-generation substrate-deficit could carry substrate-natural per-generation factor at substrate-K-type substrate-Pochhammer Sym^k:

- gen-1 V_(1/2, 1/2) substrate-Pochhammer rank substrate-natural = 2
- gen-2 V_(3/2, 1/2) substrate-Pochhammer Sym^3 substrate-natural rank · 3 = 6 = C_2 substrate-natural
- gen-3 V_(5/2, 1/2) substrate-Pochhammer Sym^5 substrate-natural rank · 5 = 10 substrate-natural

Substantive per-generation substrate-deficit RATIO V_(3/2)/V_(1/2) = 6/2 = 3 = N_c substrate-natural.

NOT 1/g = 1/7 substrate-natural directly.

4. Substantive Cat 2 (2d) Substrate-Deficit Multi-Week Path

Per Cal #189 question-shape: substrate-deficit substrate-mechanism source for 1/g substrate-natural factor not direct at substrate-Pochhammer per-generation rank substrate-natural form.

Substantive multi-week candidate: substrate-deficit substrate-mechanism per generation could enter substrate-mechanism factor 1/g at substrate-K-type substrate-Bergman + substrate-Pochhammer + substrate-deficit composite per substrate-mechanism class distinction Casey #13.

Substantive substrate-deficit per gen-2 V_(3/2, 1/2) substrate-mechanism candidate: substrate-deficit substrate-natural rank · per-Casey-#13 substrate-class factor → substrate-mechanism source for 1/g substrate-natural factor at substrate-K-type substrate-Bergman + substrate-Pochhammer composite.

Multi-week explicit derivation per Cal #189: substrate-deficit substrate-mechanism per Casey #13 + substrate-Bergman + substrate-Pochhammer composite at gen-2 V_(3/2, 1/2) substrate-mechanism class.

5. Substrate-Mechanism Source Refinement v0.1

Substantive substrate-mechanism source for Gate 4 honest open factor ~7 substrate-natural refines to:

Composite substrate-mechanism: substrate-Bergman + substrate-Pochhammer + substrate-deficit per substrate-K-type V_(3/2, 1/2) per Casey #13 substrate-mechanism class.

Substantive substrate-mechanism sources reviewed Thursday afternoon: - 4 Cat 6 substrate-symplectic walked back - 3 Cat 2 direct substrate-invariants in numerator (substrate-Pochhammer 20, $\sqrt{}$Casimir $\sqrt{}$3, substrate-Weyl 8) - 2 Cat 3 substrate-Bergman + substrate-Codim-4 substrate-natural numerator factors (225, 4) - 1 Cat 2 (2d) substrate-deficit per Casey #13 substrate-natural rank substrate-natural form (NOT directly 1/g)

Refined substantive substrate-mechanism source for Gate 4 honest open factor ~7: composite substrate-Bergman + substrate-Pochhammer + substrate-deficit per substrate-K-type per Casey #13 substrate-mechanism class; multi-week explicit substrate-mechanism composite derivation per Cal #189.

6. Closure v0.1

Cat 2 (2d) Substrate-Deficit Per Generation Candidate v0.1:

Substantive observation: substrate-deficit substrate-natural rank substrate-natural BASE constant; per-generation substrate-mechanism class distinction Casey #13 ENABLES substrate-deficit per generation distinction but substrate-deficit substrate-natural rank = 2 NOT directly 1/g substrate-natural form.

Per-generation substrate-deficit RATIO V_(3/2)/V_(1/2) = 3 = N_c substrate-natural — NUMERATOR factor not 1/g denominator.

Refined substantive substrate-mechanism source v0.1: substrate-mechanism source for Gate 4 honest open factor ~7 substrate-natural NARROWS to composite substrate-Bergman + substrate-Pochhammer + substrate-deficit per substrate-K-type V_(3/2, 1/2) per Casey #13 substrate-mechanism class; substantive multi-week explicit derivation per Cal #189.

Audit-chain Thursday afternoon substantive substrate-mechanism source candidate-narrowing CONTINUES: 4 Cat 6 walked back + 3 Cat 2 direct walked back + 3 Cat 3 numerator factors walked back + 1 Cat 2 (2d) substrate-deficit numerator factor walked back → substrate-mechanism source for 1/g substrate-natural factor REMAINS COMPOSITE multi-week per substrate-Bergman + substrate-Pochhammer + substrate-deficit per substrate-K-type substrate-mechanism category.

Per Cal #36 STANDING positive-search + Cal #27 STANDING + Cal #189: substantive Cat 2 (2d) examination → substrate-deficit substrate-natural rank substrate-base; per-generation substrate-mechanism class enables substrate-deficit per-generation distinction; substantive multi-week per Cal #189 substrate-Bergman + substrate-Pochhammer + substrate-deficit composite substrate-mechanism source.

Tier: Cat 2 (2d) substrate-deficit direct = rank substrate-natural base; substantive multi-week per Cal #189 substrate-Bergman + substrate-Pochhammer + substrate-deficit composite substrate-mechanism candidate.

— Lyra, Thu 2026-06-04 ~13:10 EDT. Cat 2 (2d) Substrate-Deficit Per Generation Candidate v0.1: substrate-deficit = rank substrate-natural base constant; per-generation substrate-mechanism class Casey #13 enables substrate-deficit per-generation distinction; substantive substrate-mechanism source for Gate 4 honest open factor ~7 NARROWS to substrate-Bergman + substrate-Pochhammer + substrate-deficit composite per substrate-K-type V_(3/2, 1/2) per Casey #13 substrate-mechanism class; substantive multi-week per Cal #189.